

Midterm Test

1. Let $A \subseteq \mathbb{R}$. Prove that $\text{int}A \subseteq A \subseteq \text{cl}A$. Give an example with strict inclusions. **1p**

2. Find the value of $\lim_{n \rightarrow \infty} \frac{1}{2n+1} + \frac{1}{2n+2} + \dots + \frac{1}{3n}$. **1p**

3. Study the convergence of the following: **3p**

(a) $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$. (b) $\int_0^{\pi} \frac{1}{\sqrt{\sin x}}$. (c) $\sum_{n=1}^{\infty} \frac{1 \cdot 3 \cdot \dots \cdot (2n-1)}{2^n n!}$.

4. Find the Taylor series around 0 and its radius of convergence for each of the following:

(a) $\ln(1+x)$. **1p** (b) $1/(1+x^2)$. **1p**

5. Consider the power series $\sum_{n=1}^{\infty} \frac{x^n}{n^2 + n}$.

(a) Find its radius of convergence and its convergence set. **1p**

(b) Find its sum for any x in the convergence set. **1p**